YAKEEN

NEET 2.0 2027

Physics | Lecture - 03

Motion in a Straight Line

By- Manish Raj (MR Sir)



Todays goal

Adv Problems on
dist^n / disp^n ✓

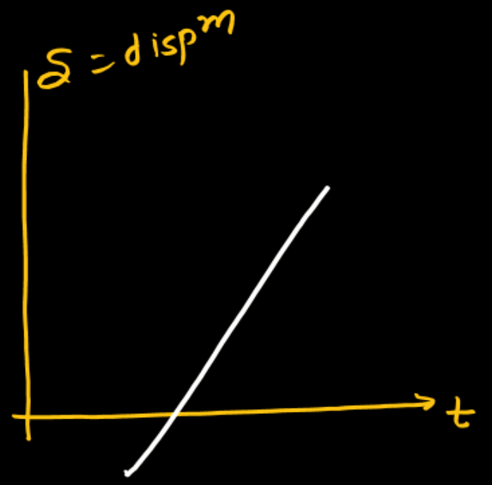
H/w goal

→ 1-D Sanghavi assignment - 1 ✓

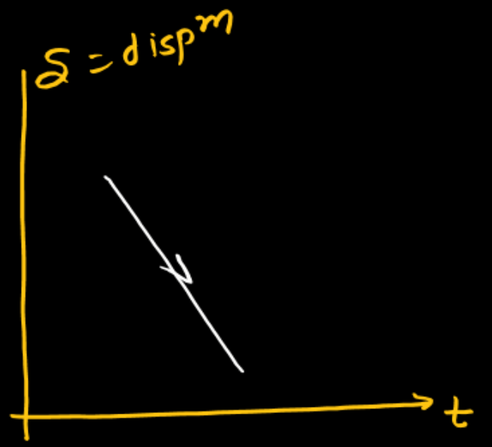
→ Revision of graph & vector addition/subtraction

MP Scam 50 Which of the following graph is Not correct:- for displacement - time

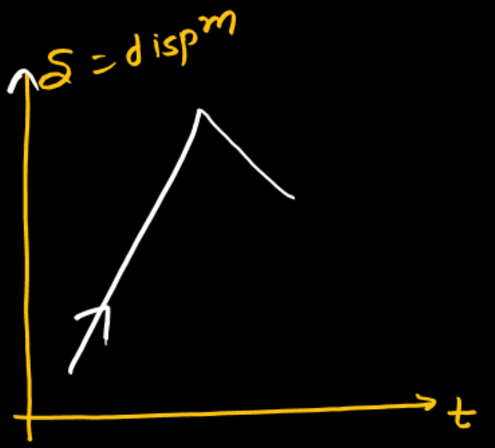
(a) Possible



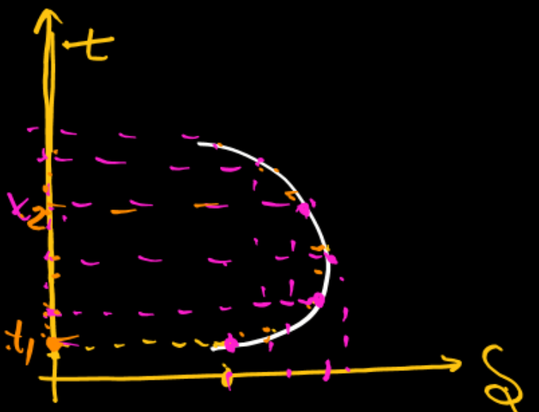
(b) Possible

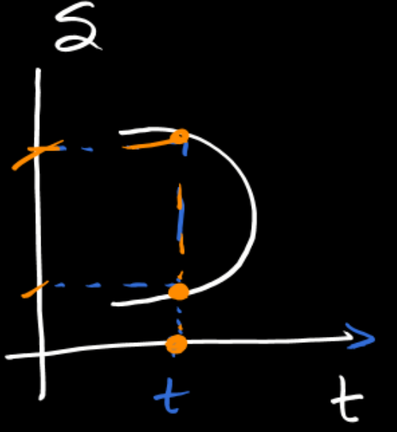


(c) Possi



(d) Possible





(e) Not Possible

Ans (e)

Question 51

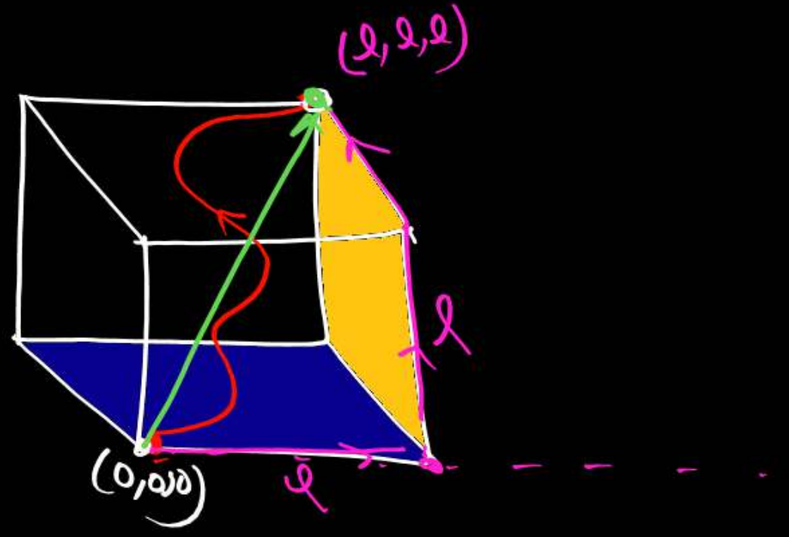
can fly

A butterfly start flying from a corner of the cubical room of side l and reaches to the opposite corner of the room. Find its displacement.

⇒ distance → Can't find

$$\text{disp}^{m} = \sqrt{(l-0)^2 + (l-0)^2 + (l-0)^2}$$

$$= \sqrt{3}\, l$$



Question

Copy

52

Can't fly

An ant start moving from a corner of the solid cubical room and want to reach opposite of body diagonal find minimum distance moved by ant. & dispⁿ

$\text{disp}^n = \ell\hat{i} + \ell\hat{j} + \ell\hat{k}$

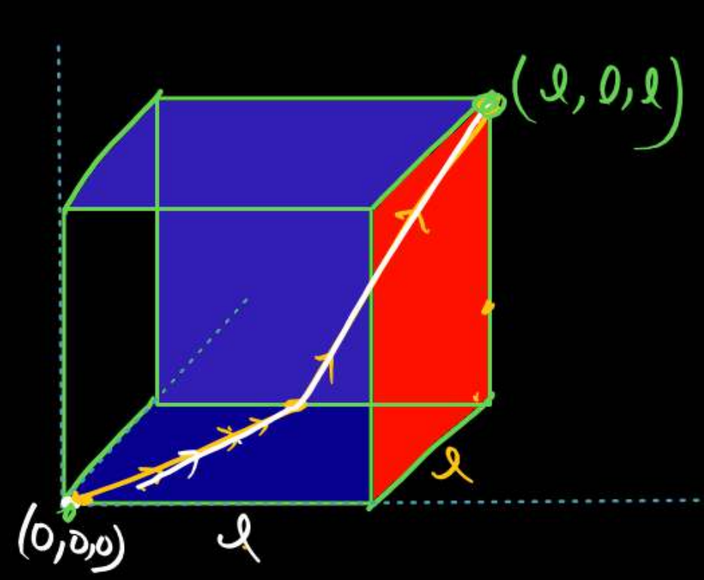
$= \sqrt{3}\ell$

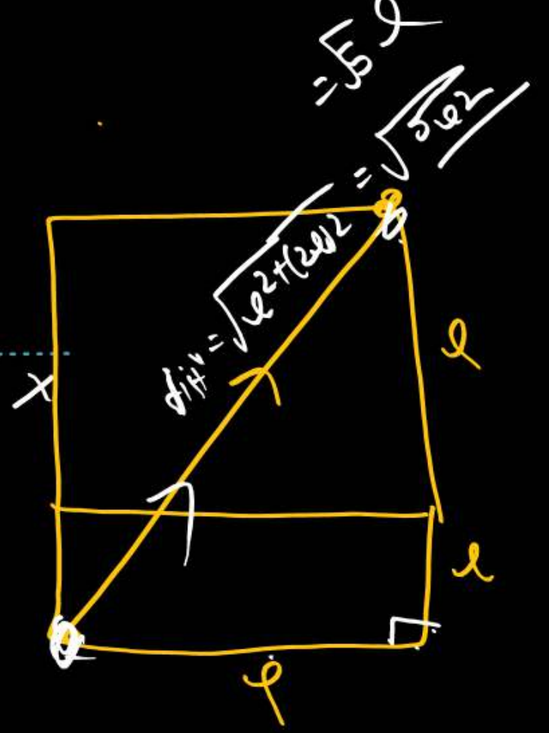
Minimum dist^n = depends on Path. $= \sqrt{3}\ell$ ✗

$= 3\ell$ ✗

$= \sqrt{2}\ell + \ell$ ✗

$= \sqrt{5}\ell$

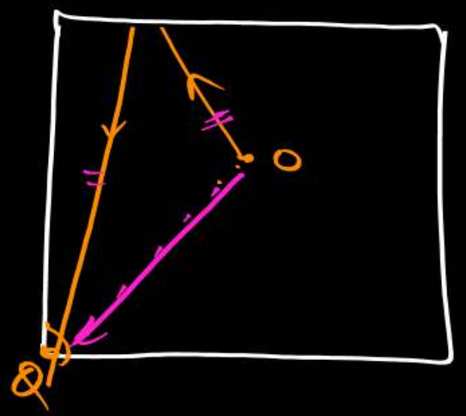


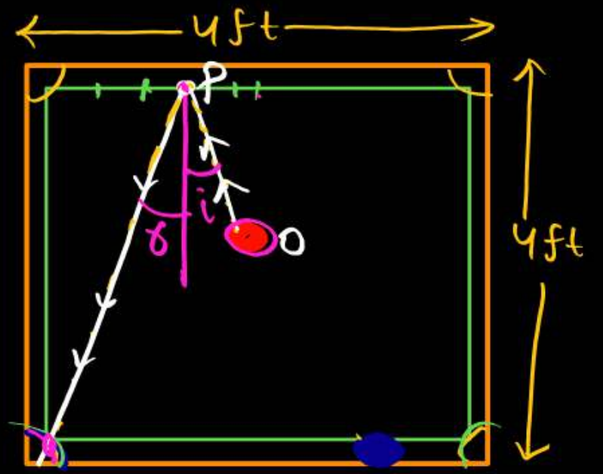


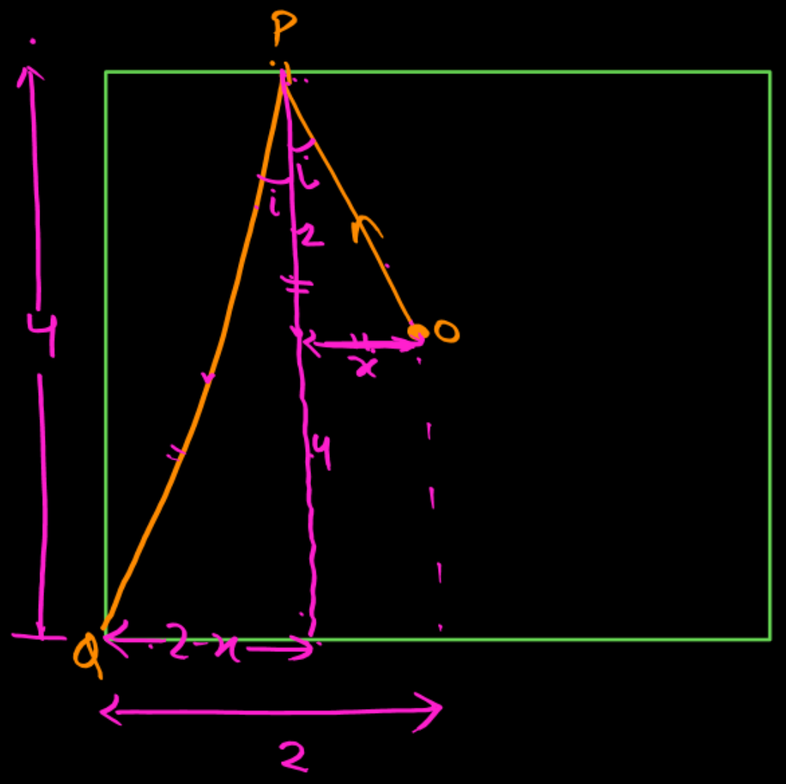
Question 53

A carrom board (4 ft × 4 ft square) has the queen at the centre. The queen, hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line. Find distance travelled & displ^{m}.

$|\vec{OQ}|$ displament $= \frac{\ell}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$







$$\tan i = \frac{x}{2} = \frac{2-x}{4/2}$$

$$2x = 2-x$$

$$3x = 2$$

$$x = 2/3 \checkmark \qquad 2-x = 2-\frac{2}{3} = \left(\frac{4}{3}\right)$$

$$(OP)_{dist} = \sqrt{\left(\frac{2}{3}\right)^2+(2)^2} = \sqrt{\frac{4}{9}+4} = \sqrt{\frac{40}{9}}$$

$$(PQ)_{dist} = \sqrt{\left(\frac{4}{3}\right)^2+(4)^2} = \sqrt{\frac{16}{9}+16}$$

$$= \sqrt{\frac{160}{9}}\ m$$

$$= \frac{\sqrt{40}}{3} + \frac{2\sqrt{40}}{3}$$

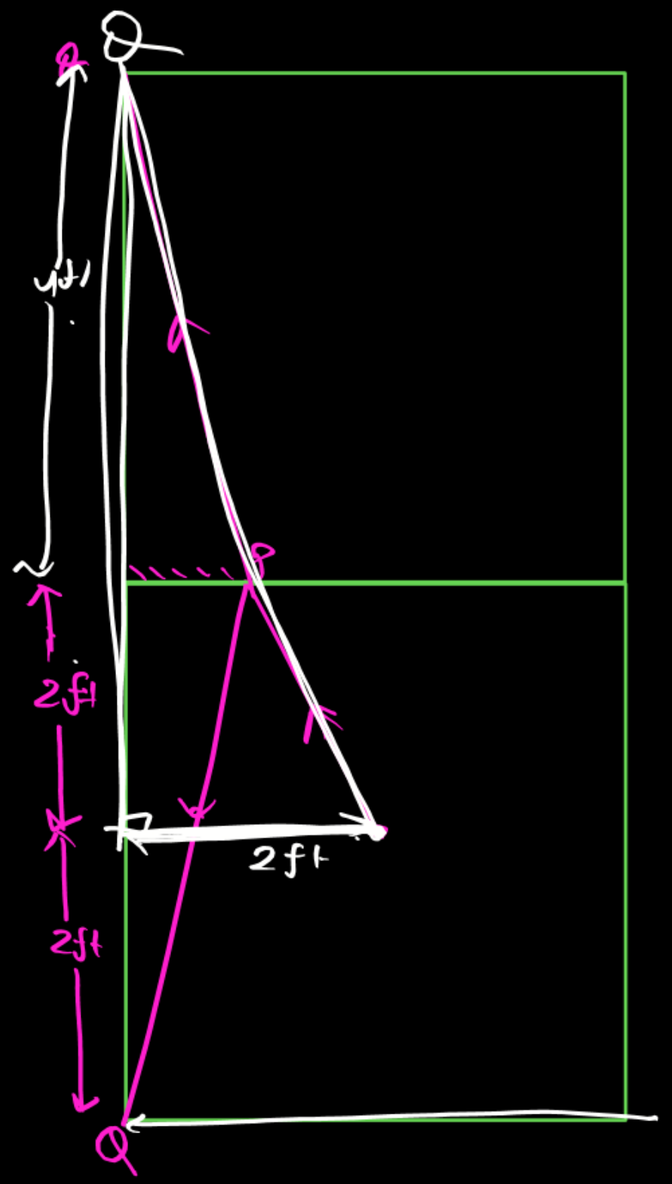
$$= \frac{3\sqrt{40}}{3} = \sqrt{40}$$

$$\text{total dist} = \sqrt{\frac{40}{9}} + \sqrt{\frac{160}{9}} \checkmark$$

$$OQ = \sqrt{(6)^2+(2)^2}$$

$$= \sqrt{36+4}$$

$$= \sqrt{40}$$



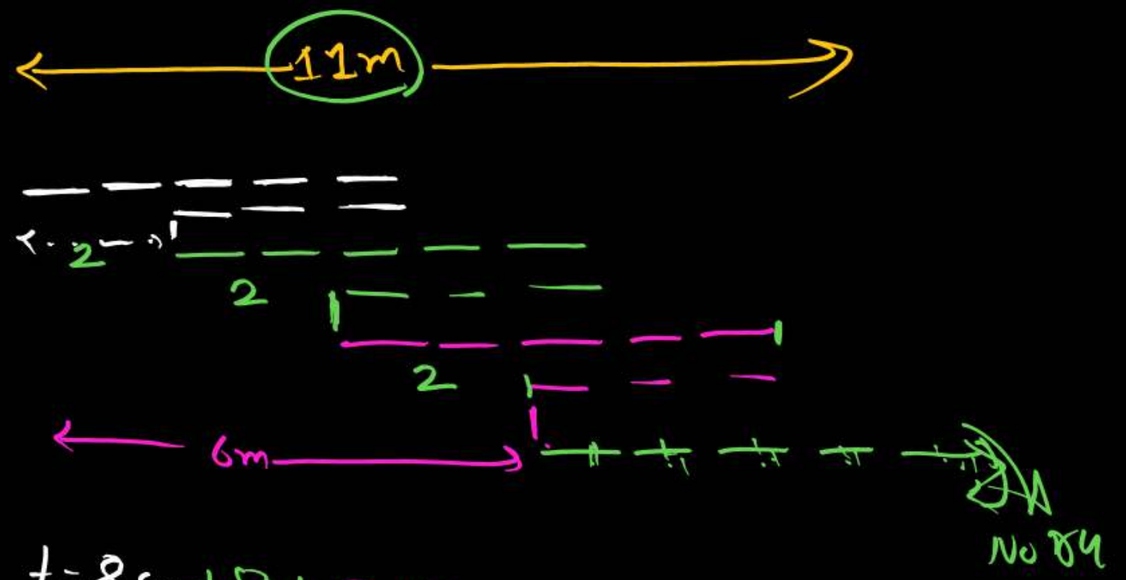
Question

51

IQ

A drunkard is walking along a straight road. He takes 5 steps forward and 3 steps backward and so on. Each step is 1 m long and takes 1s. There is a pit on the road 11 m away from the starting point. The drunkard will fall into the pit after:

1. 21 s
2. 29 s
3. 31 s
4. 37 s

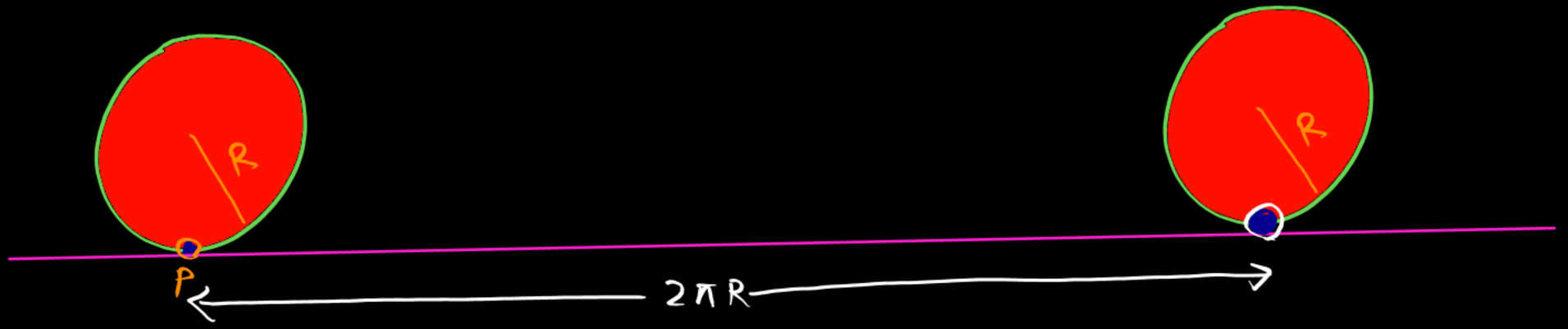


$t = 8\,sec. + 8 + 8 + 5$

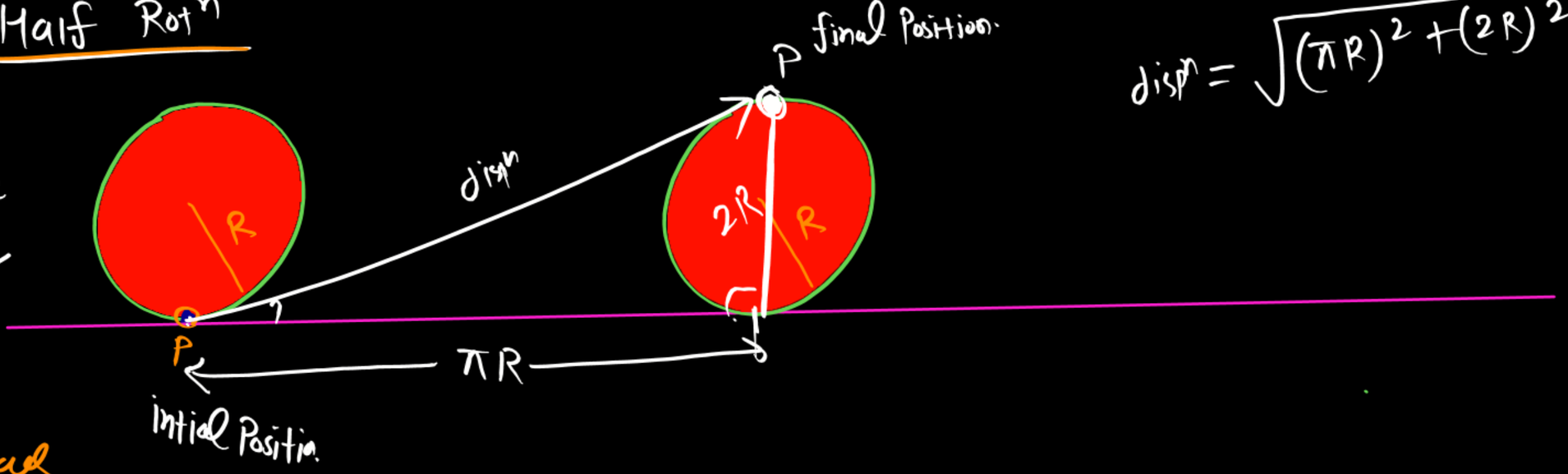
$= 29\,sec$

(55) A wheel is in Pure Rolling motion then find displacemt of Point P in

(i) full compt Rotation. $= 2\pi R$ ✓



(ii) Half Rotn



$\text{disp}^n = \sqrt{(\pi R)^2 + (2R)^2}$ ✓

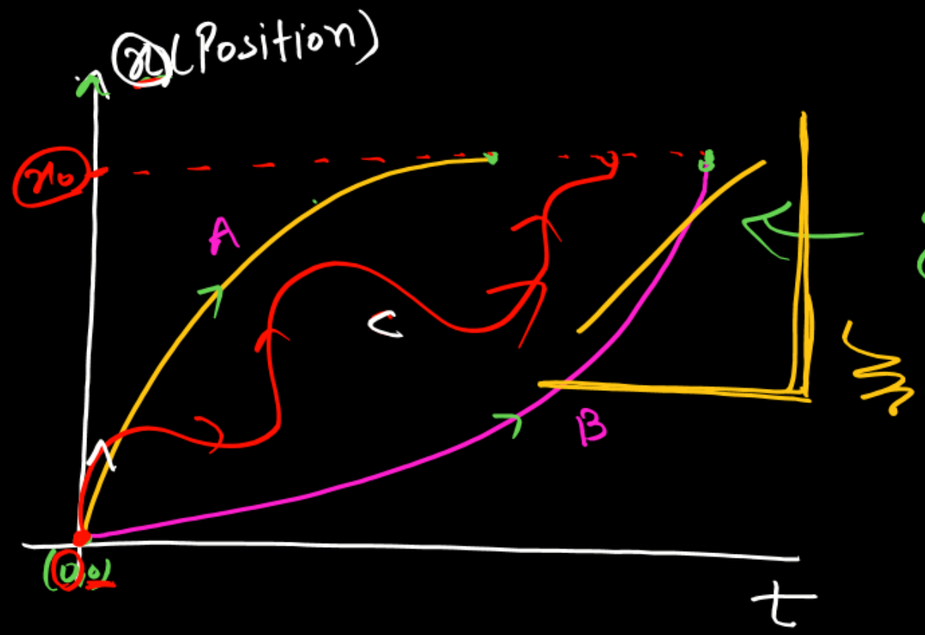
(iv) after one $\frac{1}{4}$th rotn

at glance

Not for exam

56

Which have large displacement??



graph ka length Path nahi hai

(a) A

(b) B

(c) Same for A, B & C

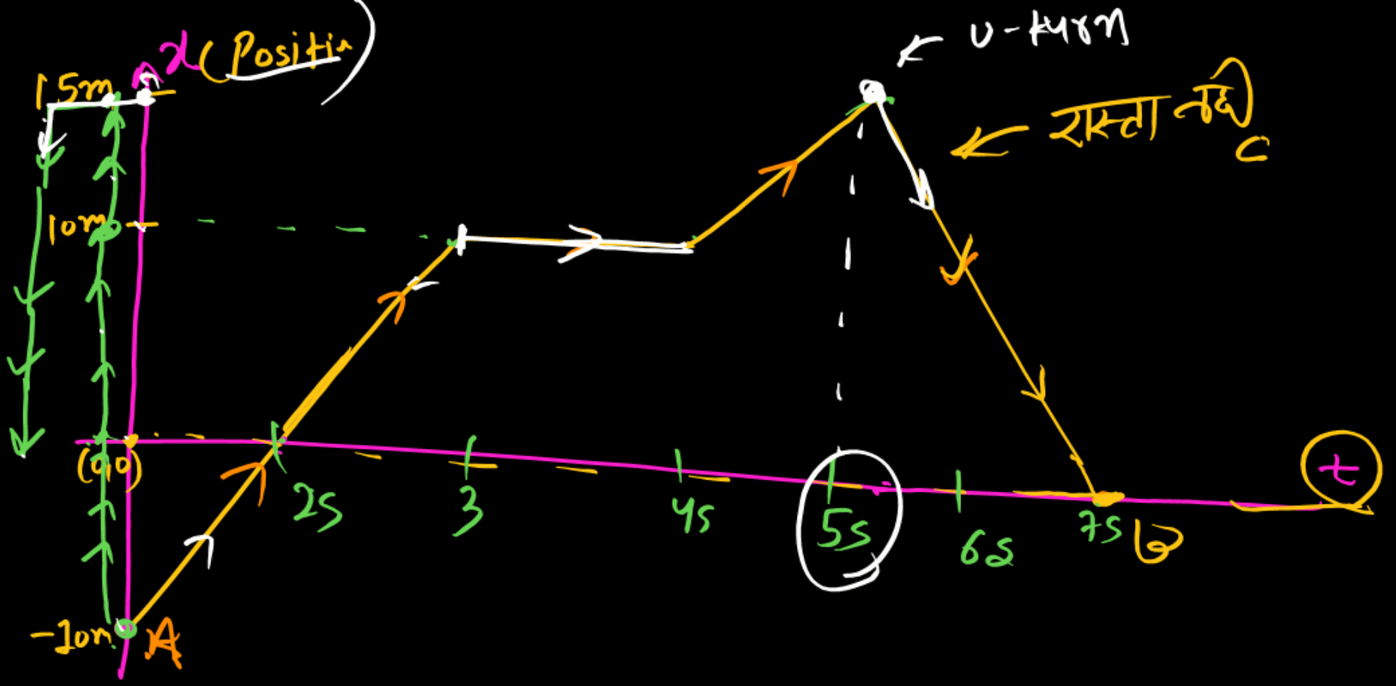
(d)

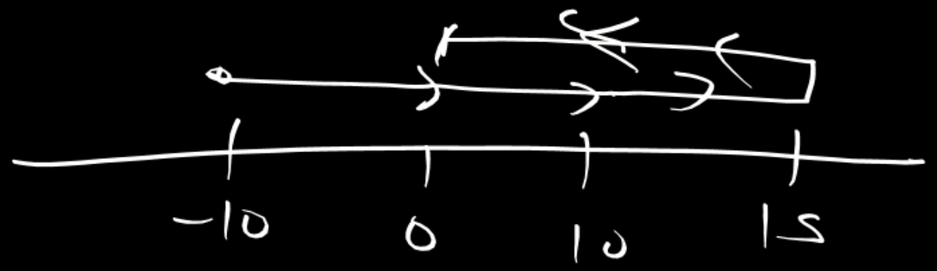
(57)

distance = 25 + 15 = 40m ✓

disp^{m} = 0 − (−10) = +10m

↳ sift initial & final Position dekho.

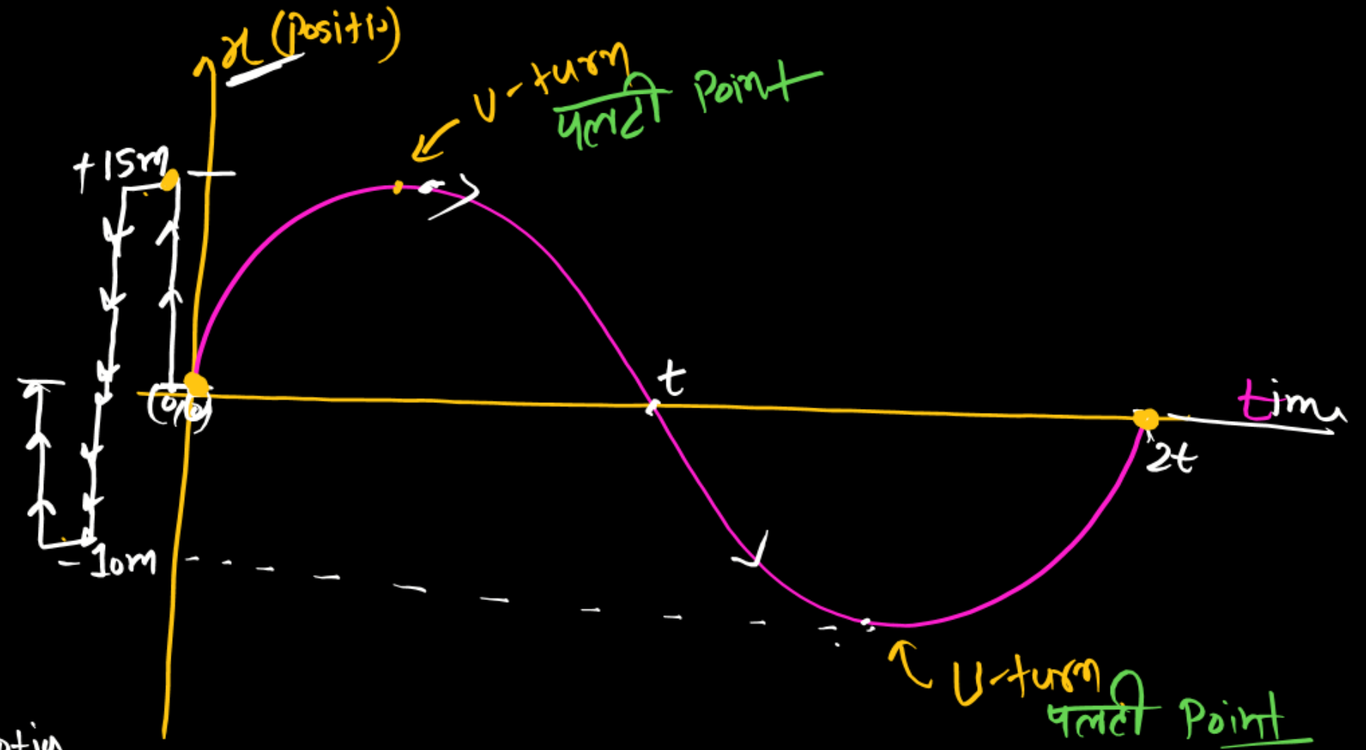




(58)

total distⁿ = 30 + 20 = 50m

total dispᵐ = 0



1-D Motion

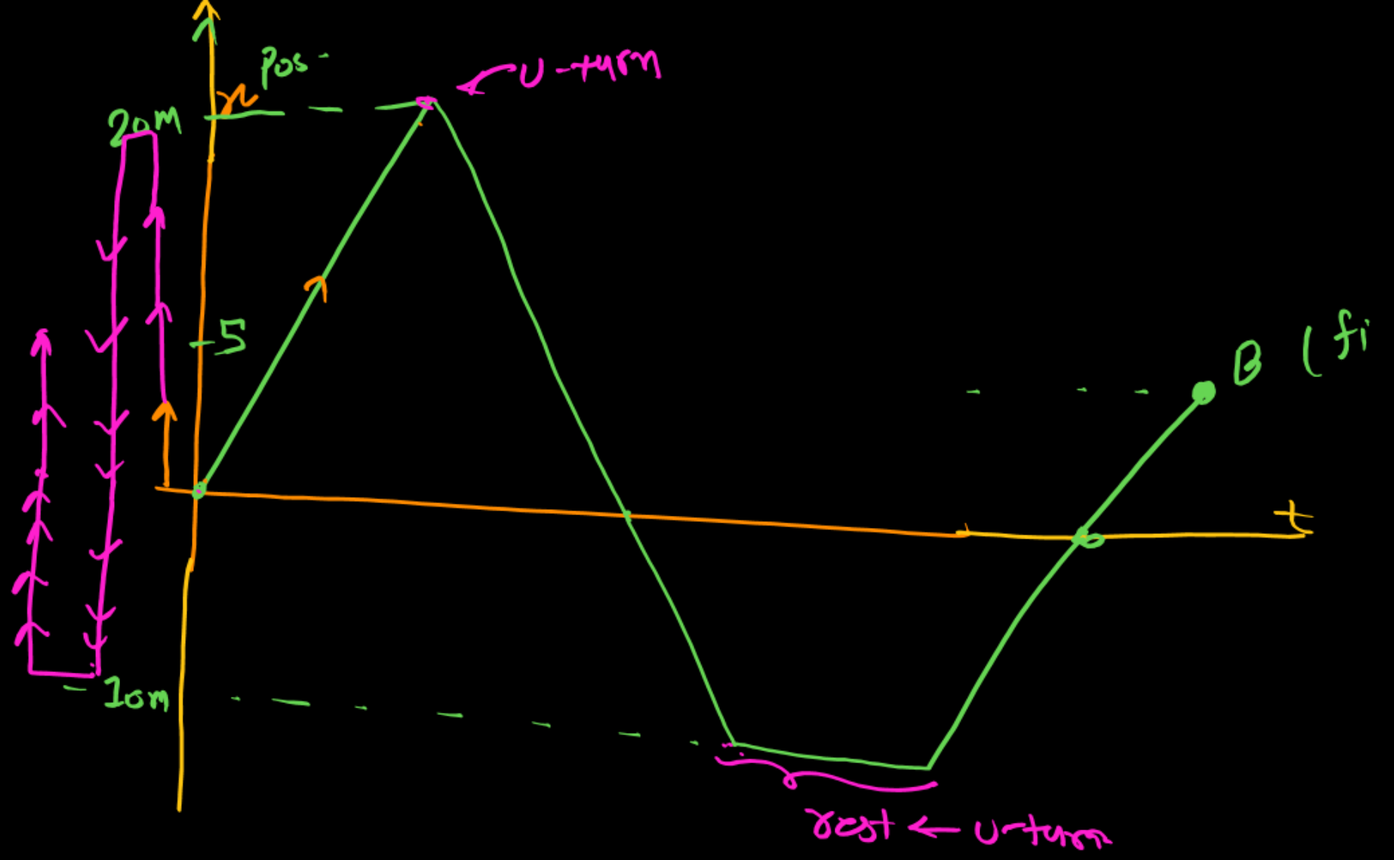
because only x-co-ordinate me Position
change ho rahi hai

59

$\text{disp}^m = 5 - 0 = 5m$

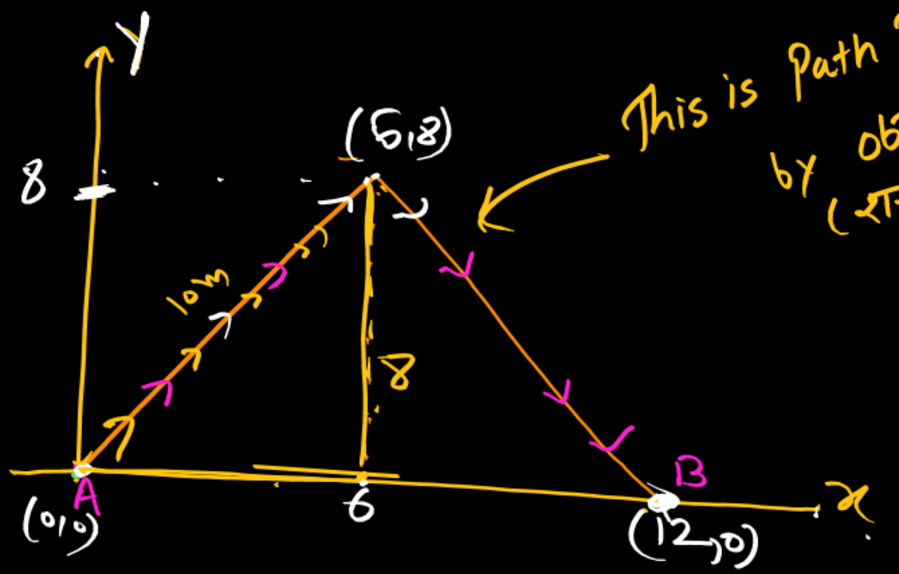
$\text{dist}^n = 20 + 20 + 10 + 10 + 5$

$= \underline{\underline{65m}}$



(60)

Find displacement from A to B ?



(a) Zero (31:1)

(b) 12m

(c) 20m

(d) 18m

distance b/w A & B

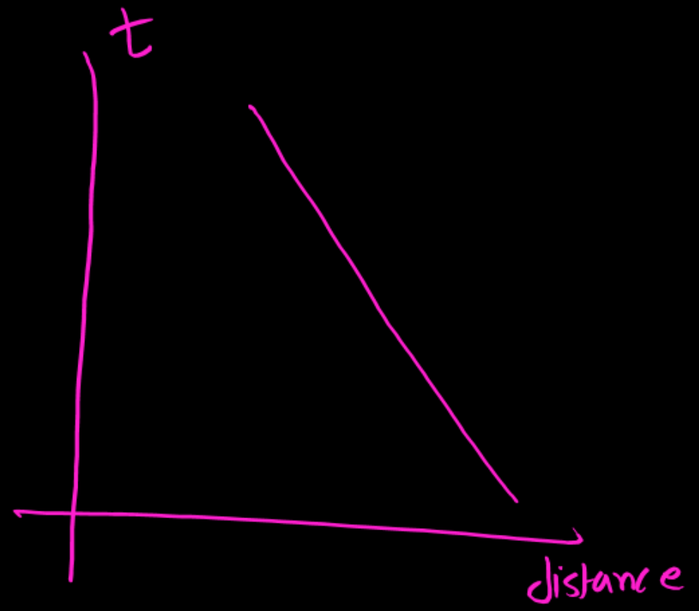
= 10+10

= 20m

$$\text{disp}^m = \vec{r}_f - \vec{r}_i$$

$$= (12\hat{i} + 0\hat{j}) - (0\hat{i} - 0\hat{j})$$

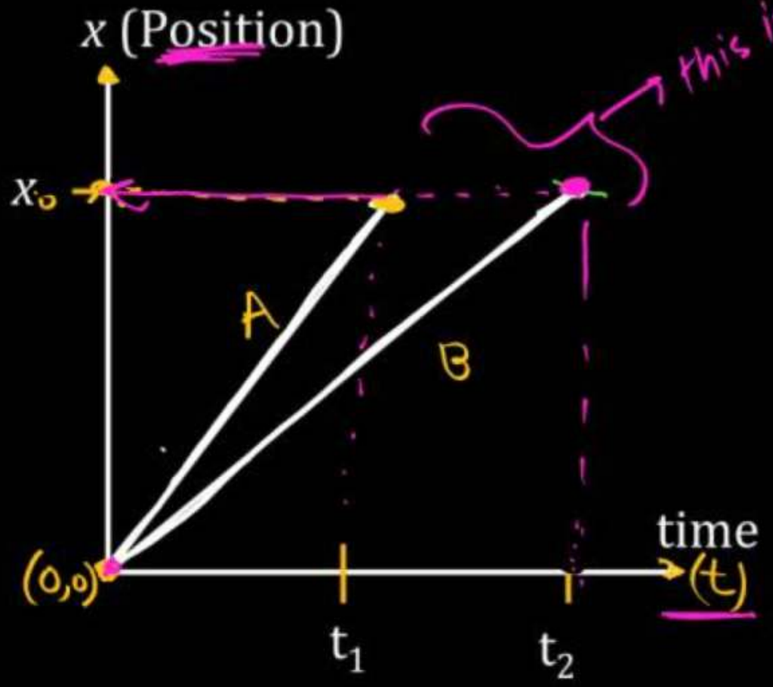
$$= 12\,m$$


t
distance

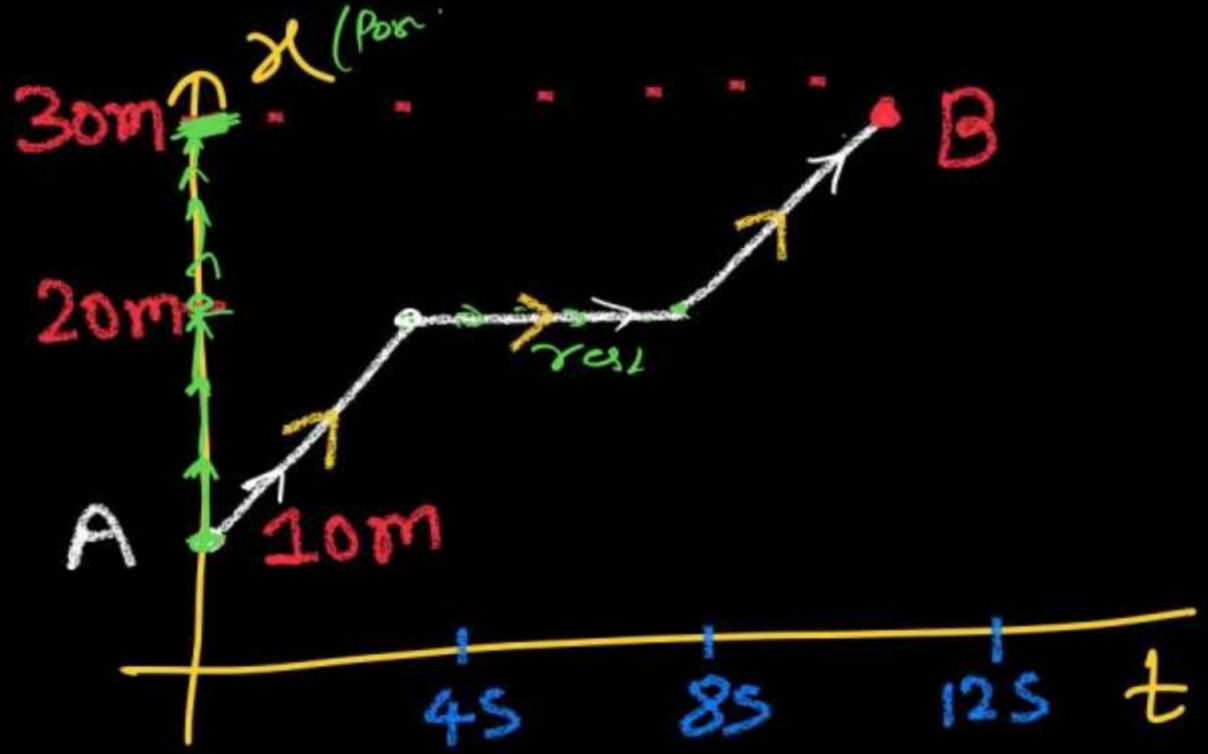
Question 42

Position-time graph for two man are moving along x-axis as shown; then find correct relation between

1. displacmeent = displament
2. displacmeent > displament
3. displacmeent < displament
4. None of these



Question 96

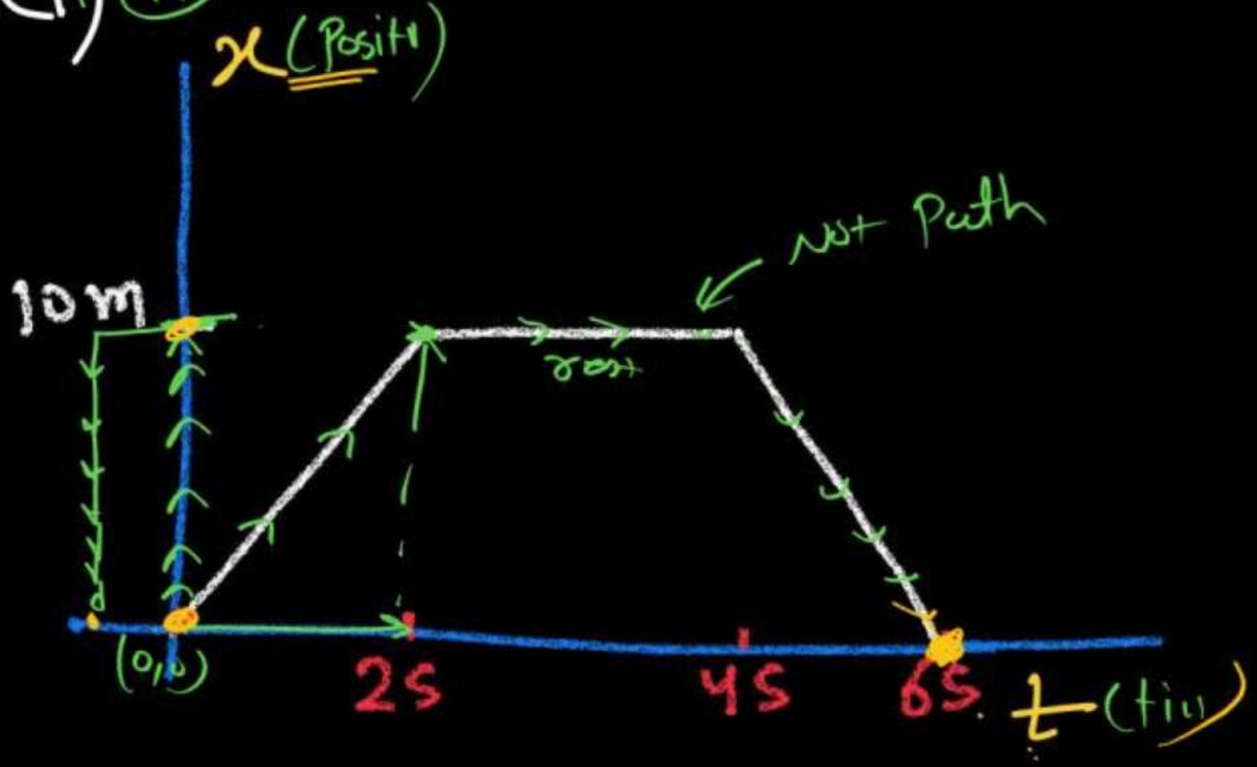


$|distance|_{A \to B} = ??$

$= 30 - 10$

$= 20\,m$

Q.14) (47)



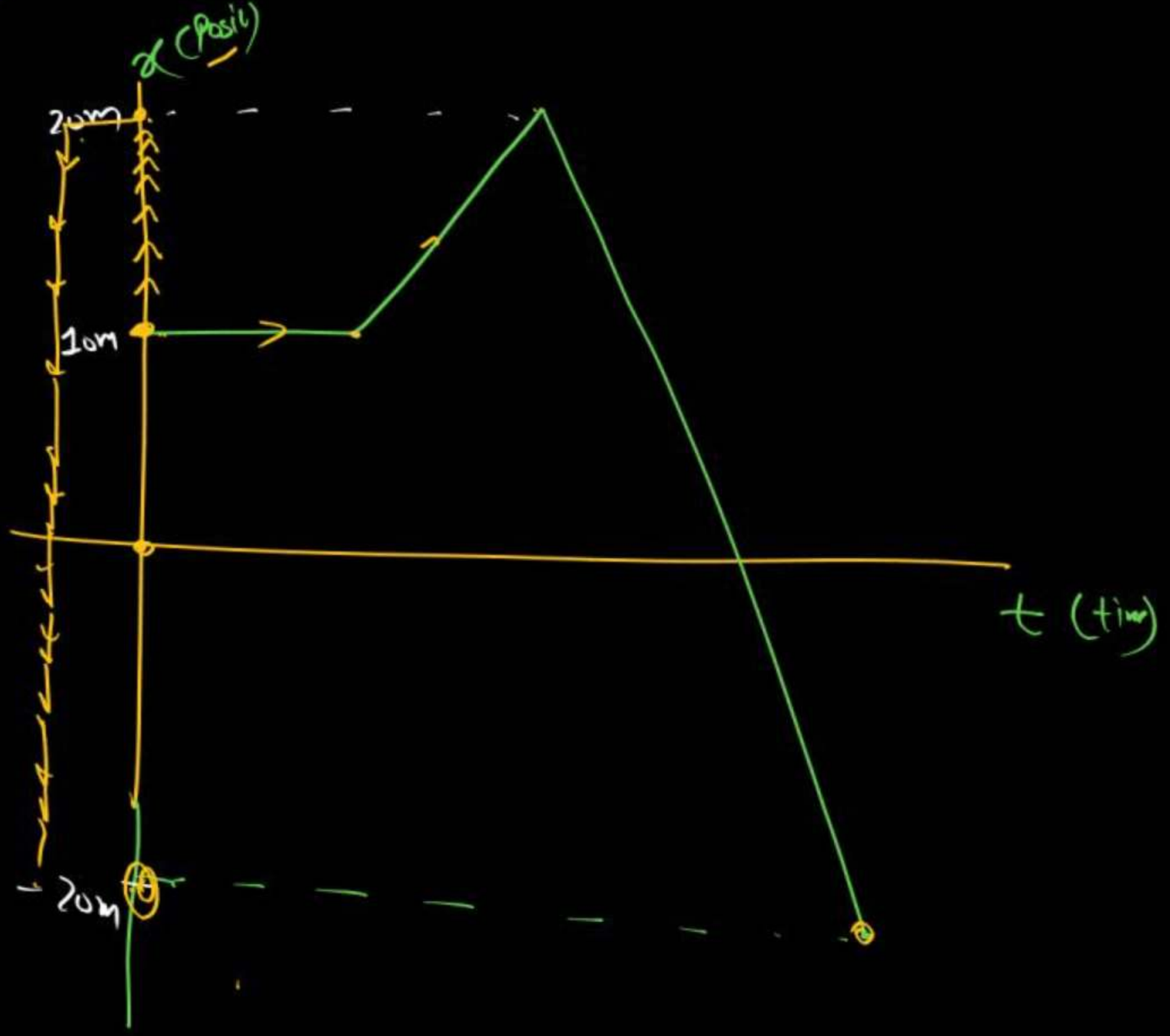
|distance|$_{A\to B}$

|displ^{m}|$_{A\to B}$

dist^{n} = 10 + 10 = 20 m

displ^{n} = 0 m

(48)



$\text{disp}^n = -20 - 10$

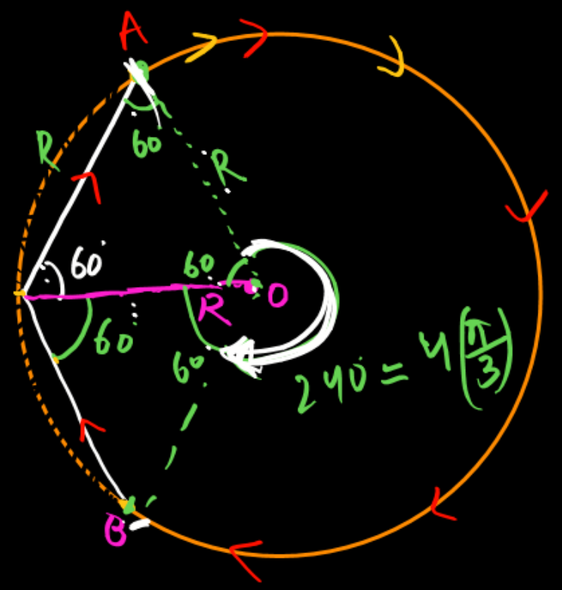
$= -30\text{m}$

$\text{dist}^r = 10 + 20 + 20$

$= \underline{50\text{m}}$

(61)

Find distance Travelled by object as shown in fig!— (R = 21cm)



distance = (Arc)$_{AB}$ + (straight line)$_{BA}$

$= R\frac{4\pi}{3} + 2R$

$= \frac{21 \times 4 \times 22}{7 \times 3} + 2 \times 21$

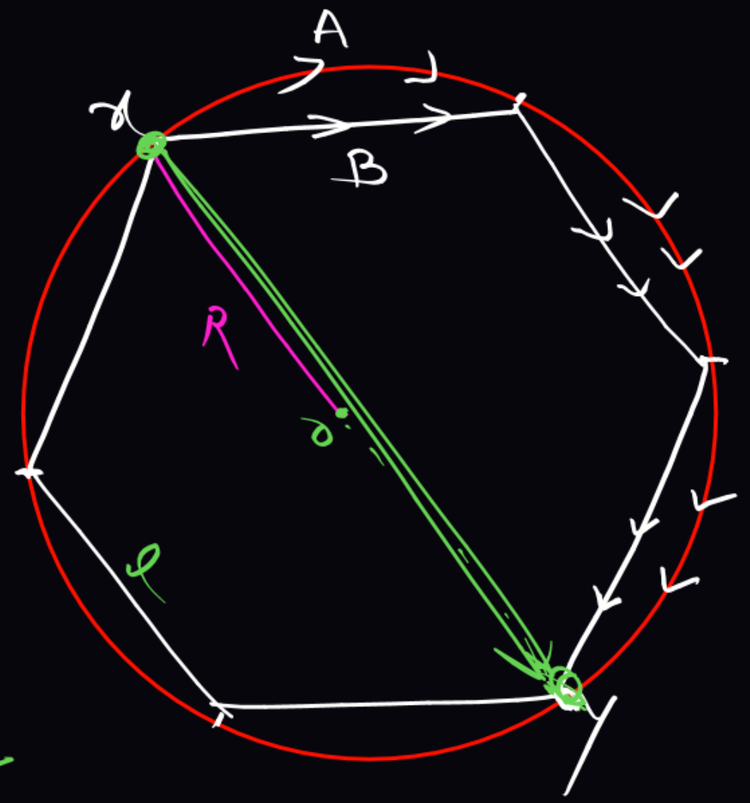
$= 22 \times 4 + 21 \times 2$

$= 88 + 42$

$= 130$

(62) A → moving on circle
B → moving on hexagon

$$\frac{|\text{disp}^m|_A}{|\text{disp}^m|_B} = \frac{1}{1}$$



$$\frac{|\text{distance}|_A}{|\text{distan}|_B} = \frac{\pi R}{3R} = \frac{\pi}{3}$$

(63)

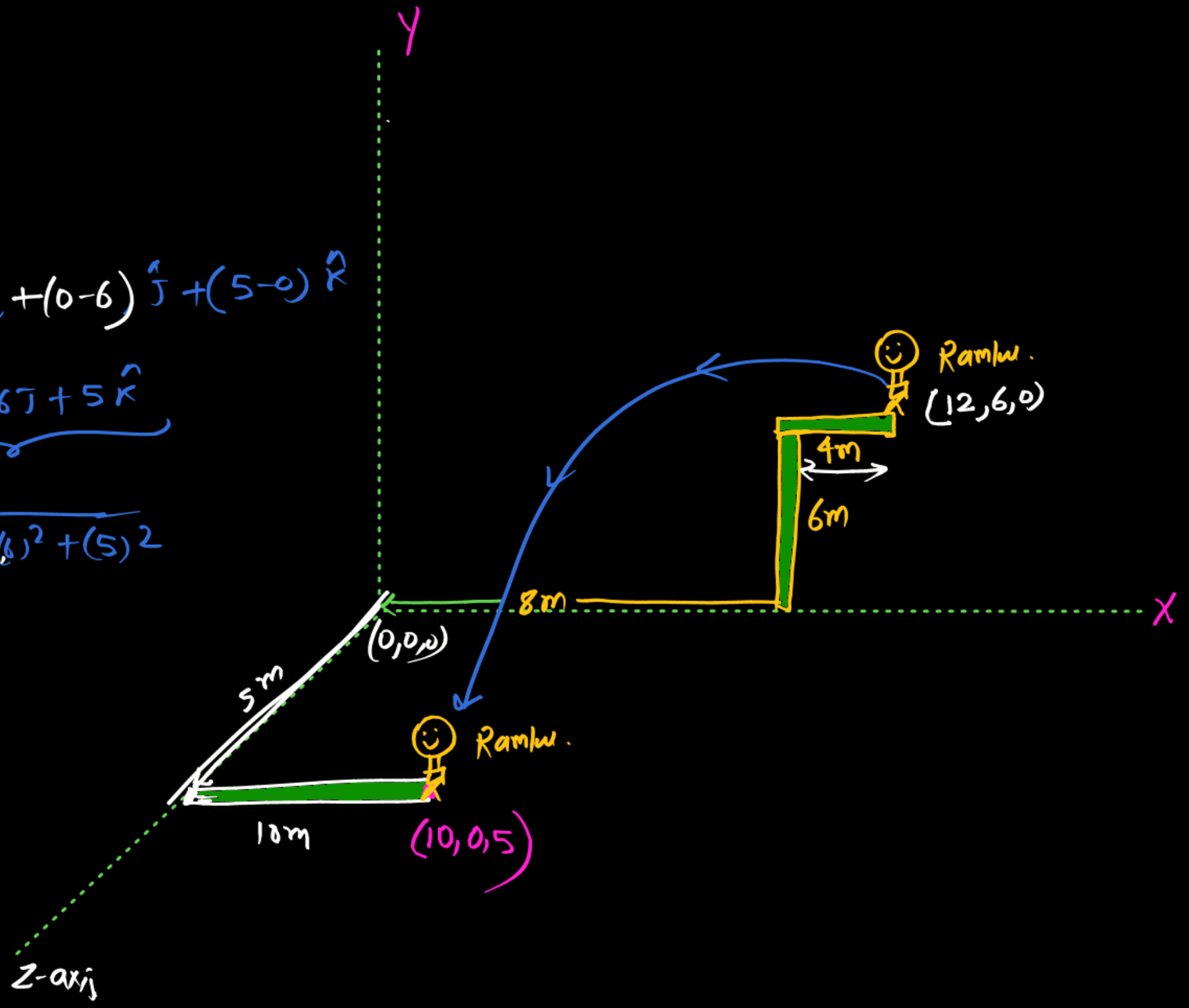
disp^m of Ramlal

$= (10-12)\hat{i} + (0-6)\hat{j} + (5-0)\hat{k}$

$= \underline{-2\hat{i} - 6\hat{j} + 5\hat{k}}$

$|\vec{s}| = \sqrt{(-2)^2 + (6)^2 + (5)^2}$

$= \underline{\sqrt{65}\ m}$



64

num/MCQ

displ^m from A to B

$$\vec{r}_B = ai + \frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}$$

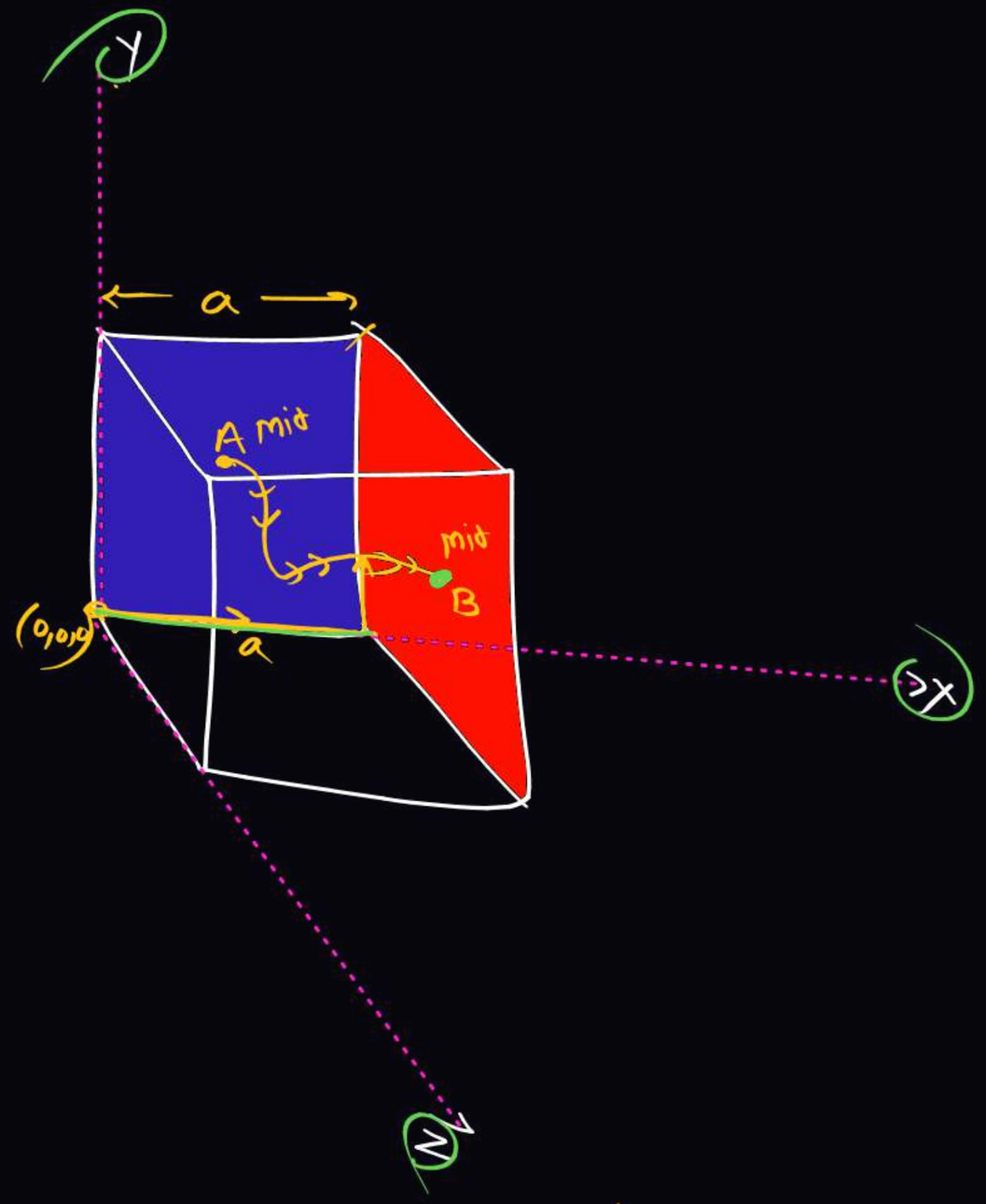
$$\vec{r}_A = \frac{a}{2}i + \frac{a}{2}\hat{j}$$

$$\text{displ}^{n} = r_B - r_A$$

$$= \left(a - \frac{a}{2}\right)i + \frac{a}{2}\hat{k}$$

$$\underline{\text{displ}^{n}} = \left(\frac{a}{2}i + \frac{a}{2}\hat{k}\right)$$

$$|\text{displ}^{n}| = \frac{\sqrt{2}a}{2\sqrt{2}} = \frac{a}{\sqrt{2}}$$ ✓



(65) Position $x = t^2 + 2t - 4$ find displacement in 2-sec.

1-D motion

U-turn hua ya nahi

at U-turn in 1-D velocity must be zero

Position at $t = 2$ sec $\quad x_f = (2)^2 + 2\times2 - 4$

$= 4 + 4 - 4 = 4m$

Position at $t = 0$ $\quad x_i = 0 + 0 - 4 = -4m$

$\text{disp}^m = x_f - x_i$

$= 4 - (-4)$

$= +8m$

$\odot$ distance $= |\text{disp}^m| = 8m$

$V = \frac{dx}{dt} = 2t + 2 - 0$

$V = 2t + 2$

If $V = 0$

$0 = 2t + 2$

$t = -1$ ✗

No turn Not Possible

66 gf Position $x = t^2 - 4t + 2$ find dist^n & disp^n in 2-sec

1-D motion

$$v = \frac{dx}{dt} = 2t - 4(1) + 0$$

$$v = 2t - 4$$

gf $v = 0$

$$0 = 2t - 4$$

$$2t = 4$$

$t = 2$ sec ← at rest & turn back

$$\Rightarrow x_{t=0} = 2$$

$$\Rightarrow x_{t=2} = (2)^2 - 4\times 2 + 2$$

$$= 4 - 8 + 2$$

$$= -2$$

$$\text{disp}^n = -2 - 2$$

$$= -4\text{m} \Rightarrow \text{distance} = +4\text{m}$$

66 gf Positn $x = t^2 - 4t + 2$ find distn & dispn in 3 sec

1-D motion

$$v = \frac{dx}{dt} = 2t - 4(1) + 0$$

$$v = 2t - 4$$

gf $v = 0$

$$0 = 2t - 4$$

$$2t = 4$$

$t = 2$ sec ✓ at rest & turn ✓

$\Rightarrow x = 2$ (initial) at $(t=0)$

$\Rightarrow x_{t=2} = (2)^2 - 4\times2 + 2$

$= 4 - 8 + 2$

$= -2$ (U-turn)

$\Rightarrow x_{(t=3)} = 3^2 - 4\times3 + 2$

$= 9 - 12 + 2$

$= -1$m final Positn



distance = +5m

dispm = −1 − (2)

= −3m

(67) $x = -t^2 + 6t + 4$ find displ^n in $t = 0$ sec to $t = 4$ sec

displ^n $= (x_f)_{t=4} - (x_i)_{t=0}$ $= 8m$ ✓

for distance (1-D), U-turn kab hoga ($V = 0$)

$V = \frac{dx}{dt} = -2t + 6 + 0$

$V = -2t + 6 = 0$

$t = 3$ sec at U-turn

$(x)_{t=0} = 4m$ ✓

$(x)_{t=3s} = -9 + 18 + 4 = 13$ ✓

$(x)_{t=4sec} = -16 + 24 + 4 = 12$



→ distance = 10m

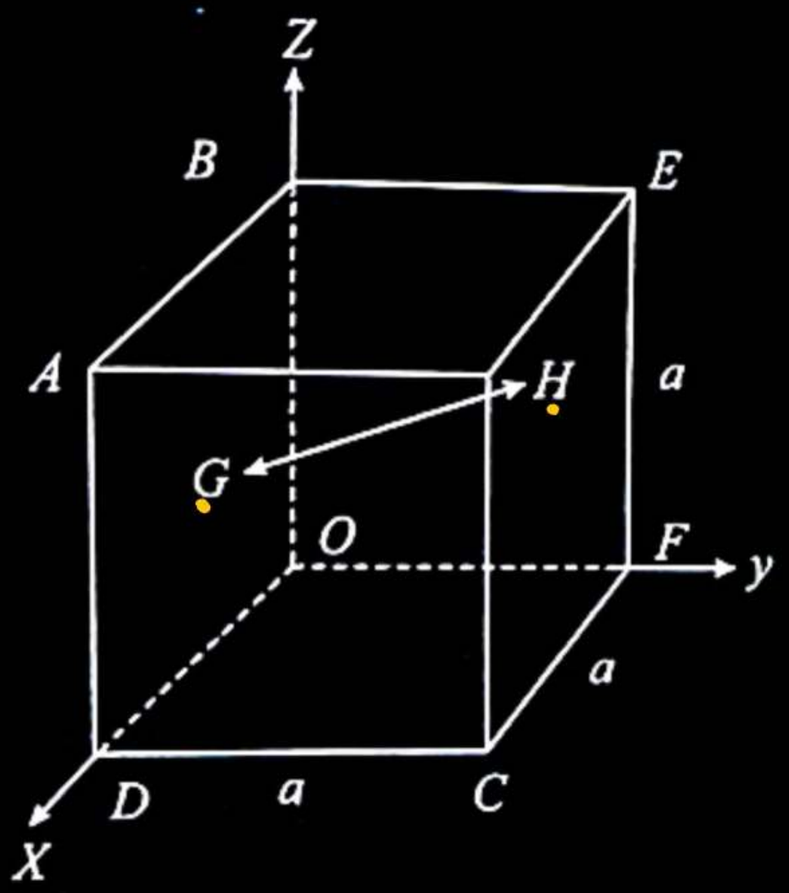
→ displ^n = 8m

Question

Not copy do at home

In the cube of side 'a' shown in the figure, the vector from the central point of the face $ABOD$ to the central point of the face $BEFO$ will be **[10 Jan, 2019 (Shift-1)]**

1. $\frac{1}{2}a(\hat{k}-\hat{i})$
2. $\frac{1}{2}a(\hat{i}-\hat{k})$
3. $\frac{1}{2}a(\hat{j}-\hat{i})$
4. $\frac{1}{2}a(\hat{j}-\hat{k})$



My **girlfriend** said she needs some **Time** and **Distance**



Is **she** Calculating **Velocity**??

Speed

On time 't'

At time 't'

$$\text{Avg speed} = \frac{\text{Total distance}}{\text{total time}}$$

= scalar

$$s = \frac{dx}{dt}$$

= slope of distance/time graph

= scalar

= m/s.

= No direction

$$\text{Avg velocity} = \frac{\text{Total disp}^{m}}{\text{total time}}$$

↳ vector

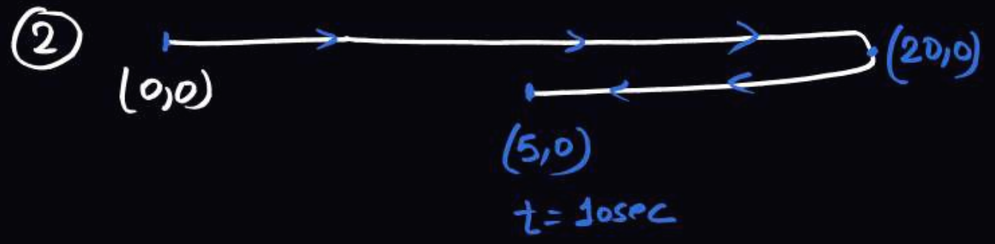
Solve in rough

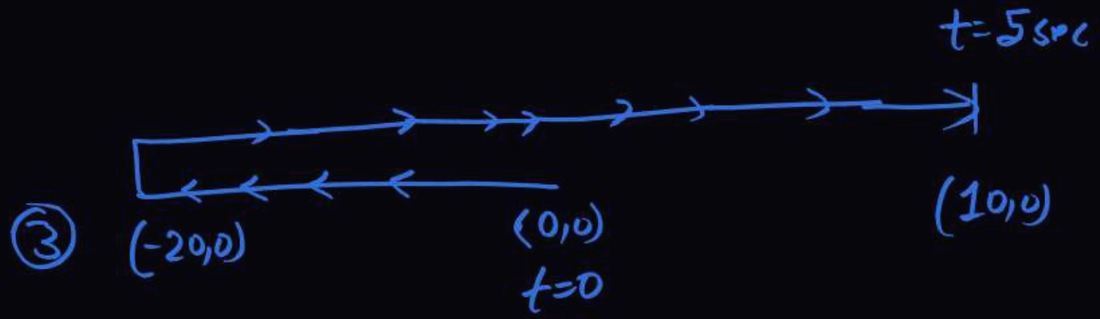
Questn 1 to 24

& copy in Notes

find Avg. speed and Avg velocity

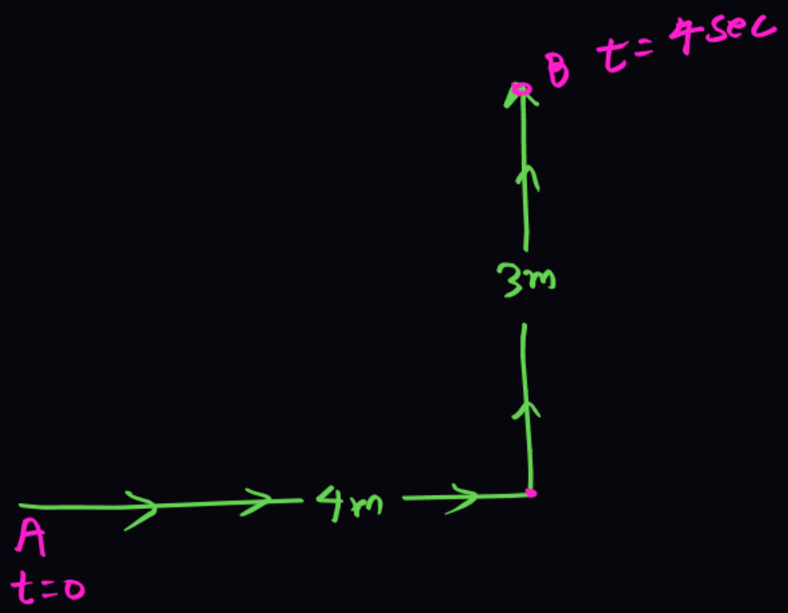




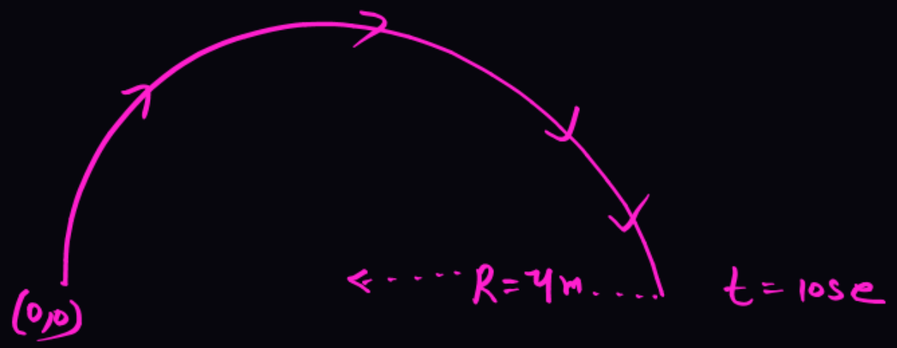




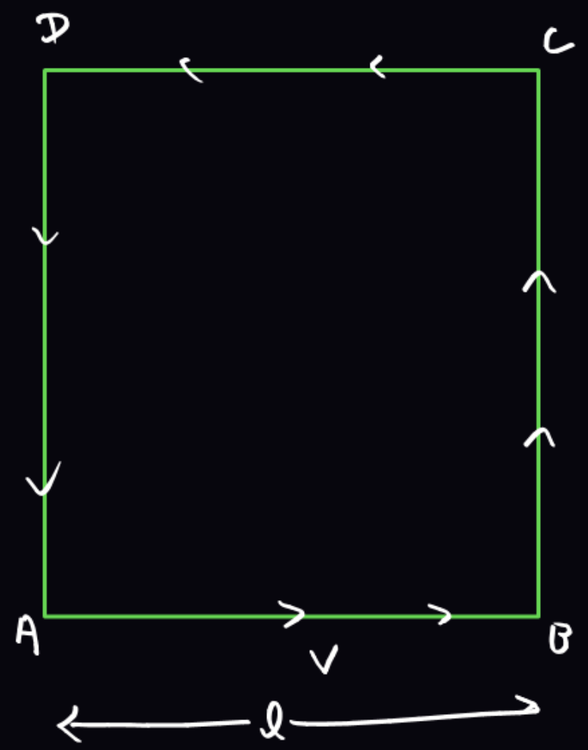
⑤



(6)



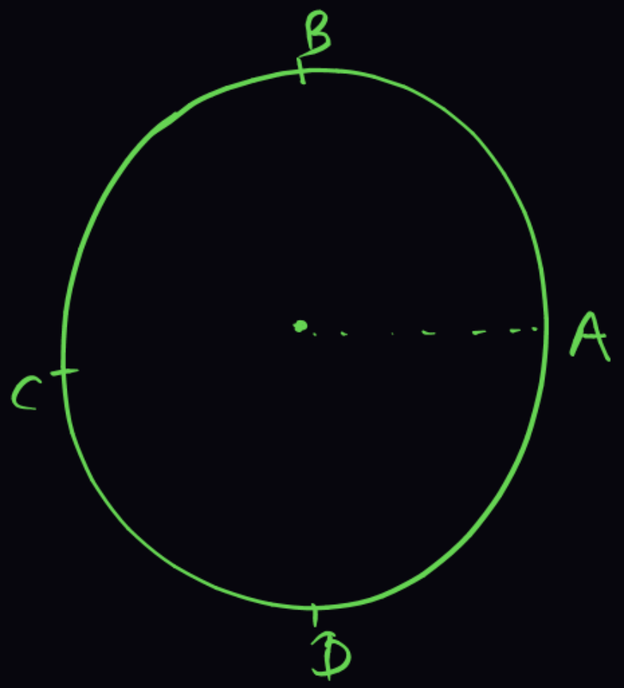
⑦ Object is moving on square park with Constant speed V then find Avg. speed & Avg Velocity for given path.



$time = \frac{disp^n}{speed}$

Motion	$dist^n$	$disp^m$	Avg Speed	Avg Velocity	time
$A \to B$					$\frac{\ell}{V}$
$A \to B \to C$					$\frac{2\ell}{V}$
$A \to D$					$\frac{3\ell}{V}$
$A \to A$					$\frac{4\ell}{V}$

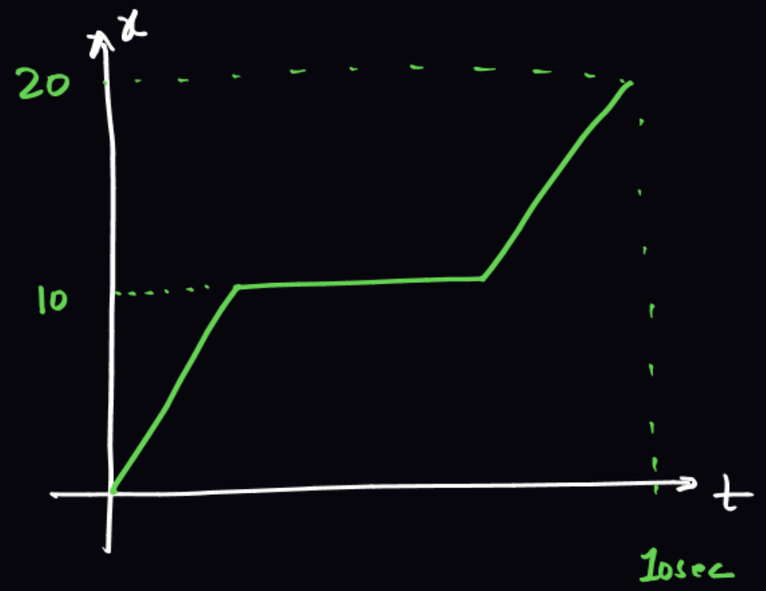
⑦ object is moving on circular Path with Constant speed V then find Avg. speed & Avg Velocity for given path.



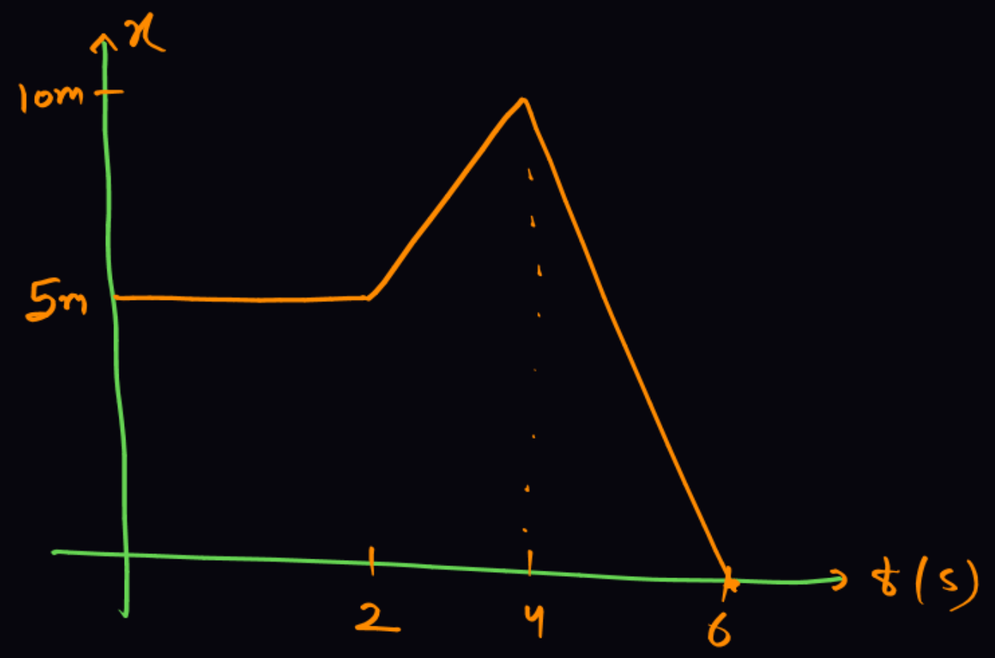
time = $\frac{\text{disty}}{\text{speed}}$

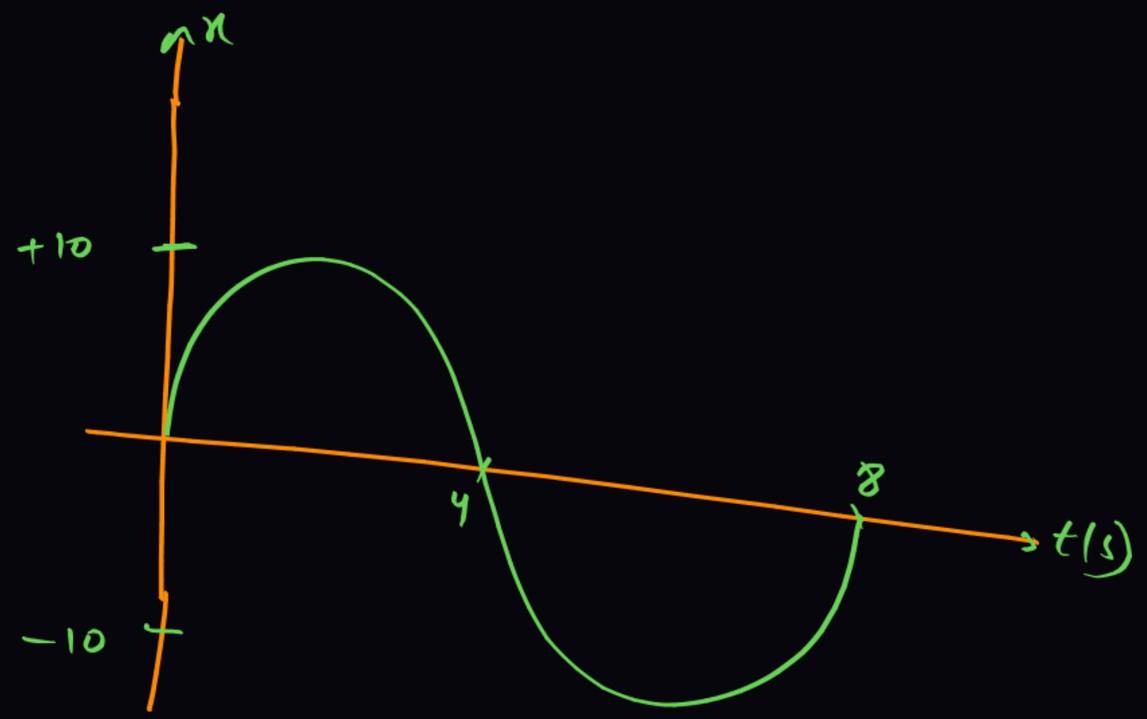
Motion	disⁿ	dispᵐ	Avg Speed	Avg Velocity	time
A → B					$\frac{\pi R}{2V}$
A → B → C					$\frac{\pi R}{V}$
A → D					$\frac{3\pi R}{2V}$
A → A					$\frac{2\pi R}{V}$

⑧



(9)





$|\text{Avg speed}|_{t=8}$

$|\text{Avg velocity}|_{t=8\,\text{sec}}$

(17) Object moves 10m in 2 sec then in same dirn 20m in 3 sec

find Avg speed and velocity.

(12) object moves with speed s_1 for time t_1 then speed s_2 for time t_2 then Avg speed.

(13). object moves with speed s_1 for distance x_1 then speed s_2 for distance x_2 then Avg speed.

Question

14

A vehicle travel half the distance with v and remaining half with $2v$ then average speed?

1. $v/3$
2. $4v/3$
3. $2v/3$
4. $3v/4$

Question

15

A train has speed 60 km/hr for one hour and 40 km/hr for next half hour, then average speed

1. 50
2. 53.33
3. 48
4. 70

Question

16

Object moves for 10s with speed 20 m/s and then it moves with speed 30 m/s for next 10 sec. Then find average speed for complete journey?

Question

17

Object is moving with speed ‘S’ on square of side *l* then find average speed and average velocity for given motion

Question

18

Object moves 10m with speed 20 m/s and then it moves with speed 30 m/s for 10 sec. Then find average speed for complete journey?

Question

19

Object moves 20m with speed 20 m/s and then it moves with speed 30 m/s for 20m. Then find average speed for complete journey?

Question

20

If object moves $\frac{2}{5}^{th}$ distance of journey with speed 10 m/s and remaining with 30 m/s then average speed will be :

Question

21

A car travels from Kota to Jaipur with speed 30 km/h, and it returns along the same path with speed 60 km/h. Calculate average speed of the car.

Question

22

A body covers first one-third of the distance with velocity 10 ms^{-1} in same direction, the second one-third with a velocity 20 ms^{-1} and last one-third with a velocity of 30 ms^{-1}. The average velocity of body is

1. 17.8 ms^{-1}
2. 16.4 ms^{-1}
3. 18.3 ms^{-1}
4. 20.2 ms^{-1}

Question 23

Object moves with v_1 for $t/3$ and with v_2 for $2t/3$ then find average speed.

Question 24

A bus travels its half distance of journey with speed 5 m/s. It covers remaining distance in two equal time intervals with speed 15 m/s. Calculate average speed of the bus for the whole journey.

Thank

You